3 **(a)** PQRS comprises of a rough slope PQR of height 1.1 m, and smooth horizontal surface RS. PQR of total surface length 2.7 m is made up of a straight section PQ and curved section QR as shown in Fig. 3.1.

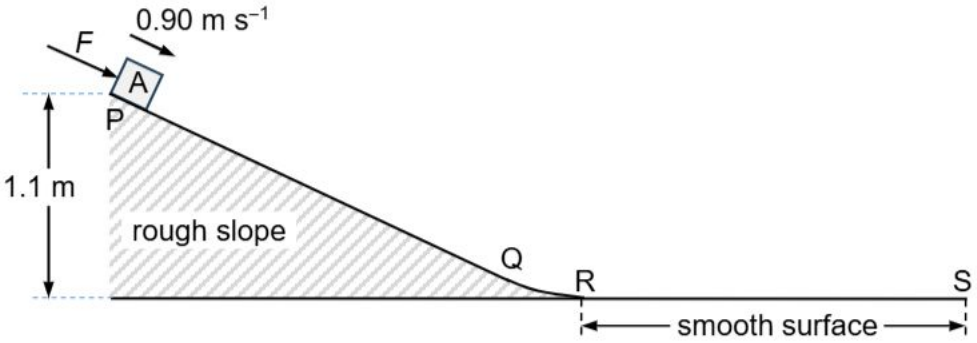


Fig. 3.1

Block A of mass 2.0 kg has an initial speed of 0.90 m s^{-1} at point P on rough slope. A force F is applied to block A as it moves along PQR. The magnitude of the F is 11 N and its direction is the same as the object's velocity at any instant along surface PQR.

The frictional force between the block A and the surface PQR may be taken to be constant at 6.4 N.

(i) Determine the speed of the block when it reaches R.

gain in KE = WD by gravitational force + WD by F − WD by friction

$= mgh + F.s - F_{friction}.s$

$= (2.0)(9.81)(1.1) + (11)(2.7) - (6.4)(2.7)$ [B1]

$= 34.002 = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$

$v = \sqrt{\frac{2(34.002)}{m} + u^2}$

$= \sqrt{\frac{(2)(34.002)}{2.0} + (0.90)^2}$ [B1]

$= 5.9$ m s^{-1} [A1]

speed = ... m s^{-1} [3]

(ii) Use Newton's second law to explain how the speed of block A varies along PQRS.

Along PQ the net force given by (F + component of weight parallel to slope $W_{//}$ – friction) = ma. (acceleration is constant and) Speed increases as $F_{net} > 0$ (as F > friction).

For the section QR, as $W_{//}$ decreases to zero as the slope flattens out, F_{net} is still > 0, speed of block increases further (at a decreasing rate).

(F_{net} goes from (F + component of weight parallel to slope $W_{//}$ – friction) to (F – friction)

For RS, the block moves at constant speed as F_{net} = 0, no applied force and no friction between surfaces.

[3]

(b) Block X of mass 0.27 kg moves with a speed 0.67 m s^{-1} on a smooth horizontal surface. Block Y of mass 0.34 kg moves with speed 0.46 m s^{-1} in the opposite direction. A light spring of natural length 13.0 cm and force constant 120 N m^{-1} is attached to block Y as shown in Fig. 3.2.

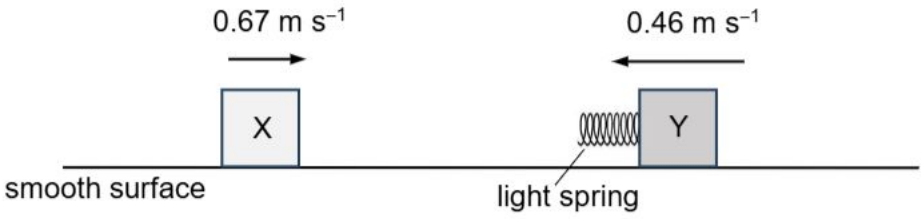


Fig. 3.2

When block X collides elastically head-on with block Y, there will be an instant when the spring experiences maximum compression.

Determine the closest distance of approach of the two blocks.

At closest distance of approach, both block moves with the same speed. [C1]
Taking rightwards as positive

$$m_A u_A + m_B u_B = (m_A + m_B)v$$

$$v = \frac{m_A u_A + m_B u_B}{m_A + m_B} = \frac{(0.27)(0.67) + (0.34)(-0.46)}{0.27 + 0.34} \qquad \text{[C1]}$$

$$= 0.0402 \text{ m s}^{-1}$$

Difference in KE before collision and max compression = EPE stored in spring

$$\left(\frac{1}{2}m_A u_A^2 + \frac{1}{2}m_B u_B^2\right) - \frac{1}{2}(m_A + m_B)v^2 = \frac{1}{2}kx^2$$

$$\left(\frac{1}{2}(0.27)(0.67)^2 + \frac{1}{2}(0.34)(-0.46)^2\right) - \frac{1}{2}(0.27 + 0.34)(0.0402)^2 = \frac{1}{2}(120)x^2 \qquad \text{[C1]}$$

$$x = 0.0400 \text{ m}$$

closest distance of approach $= L_0 - x$

$= 0.130 - 0.040 = 0.090$ m [A1]

closest distance of approach = ... m [4]

[Total: 10]